Course Syllabus

Course Code	CPE231
Course Name	Algorithms
Course Credits	3 (3-0-6)
Semester/Year	1/2026
Prerequisite	Discrete Mathematics, Programming with Data structures
Class meeting	Tuesday 13.30-16.20 Room 1115
Class website	https://www.leb2.org
Course Instructor	Assoc.Prof.Dr. Peerapon Siripongwutikorn **Email**: peerapon.sir@kmutt.ac.th **Facebook**: Peerapon Siripong
Office hours	Thursday 13.30-15.00 Room 1038 or By appointment
Teaching Assistant	Mr. Thanapat Thaiking **Email**: thanapat.thaik@kmutt.ac.th **Facebook**: Thanapat Thaiking
Course Description	The course introduces students to algorithm design strategies and solutions for common types of algorithmic problems including sorting, graph problems, network flow, matching, dynamic programming, and combinatorial search and optimization, and familiarity with an assortment of important algorithms. By the end of the course, students should be able to explain most of key algorithms, their design strategy, and their efficiency in solving problems. Students should recognize basic well-known problems and articulate whether they are easy or hard, compare and select appropriate algorithms, with good justification, to work out the problem solutions.
Course Learning Outcomes (CLOs)	After completing this course, students should be able to CLO1. Demonstrate mastery of principles and concepts in algorithm analysis and design. CLO2. Apply suitable algorithms to solve basic computing problems. CLO3. Analyze a problem and mathematically formulate the algorithmic solution to solve it.
Teaching Method	Lecture, In-class hands-on activities
Student Evaluation	In-class assignments 15% Homework 20% Exam 1 30% Exam 2 35% A: ≥ 80, B+: [84,80], B: [79,75], C+: [74,65], C: [64,55], D+: [54,50], D: [49,45], F: < 45 The instructor reserves the right to change the grading policy as deemed appropriate.
Key references	▷ T.H. Cormen, et.al, Introduction to Algorithms, 4th ed., MIT Press, 2022 ▷ A. Levitin, Introduction to the Design and Analysis of Algorithms, 3rd ed., Pearson Education, 2012. ▷ J. Kleinberg and E. Tardos, Algorithm Design, Addison-Wesley, 2006
Class policies	**Late submissions** are only accepted under reasonable excuses and explicit permission from the instructor, or it will be deducted one grade off each day. No submission is accepted after the solution has been posted. **Posted solutions** will be brief and does not show routine works. You should attempt to work out detailed solutions on your own. **Academic integrity** is strictly enforced.

Course Schedule

The following topics will be covered in our schedule. The schedule may slightly change, depending on our progress in the class.

Date	Topics
Aug 4	Introduction
Aug 11	Algorithm efficiency analysis
Aug 18	Linear time sorting and order statistics
Aug 25	Graph problems – DFS review, Topological sorting, Critical path, Articulation points, Strongly connected components, Negative weight graph and cycle; All-pair shortest paths
Sep 1	Network flow – Max-flow problem, Ford-Fulkerson method, Cut and cut capacity, Min-cut max-flow theorem
Sep 8	Network flow applications – Network connectivity, Project selection, Circulation problem
Sep 15	Matching in Bipartite graph (1) – Maximum matching, Hopcroft-Karp algorithm, Stable marriage and college admissions problems.
Sep 22	Matching in Bipartite graph (2) – Maximum weighted matching, One-to-many maximum matching
Sep 29	String matching (1) – Brute-force technique, Horspool's and Boyer-Moore algorithms
Oct 6	String matching (2) – Finite Automaton, Rabin-Karp finger printing
Oct 13	*No class – Nawamindra Maharaj Day*
Oct 14-28	**University Exam Period (Exam 1)**
Nov 3	Dynamic programming – Basic concepts and problems; Interval scheduling, Knapsack problem;
Nov 10	Sequence alignment and similarity (1) – Alignment and edit distance, Longest common subsequence
Nov 17	Sequence alignment and similarity (2) – Minimum cost alignment, Levenshtein distance
Nov 24	Combinatorial search and optimization – Easy and hard problems, Backtracking, Branch-and-Bound, Heuristics
Nov 30 - Dec 11	**University Exam Period (Exam 2)**

Notes:

- Exam 1 covers the materials in the first six weeks of the above schedule (Up to *Network flow applications*), and Exam 2 covers the remaining materials.
- Check the university exam schedule for the exam dates.